

Stochastic Storm Surge Modeling using ADCIRC+SWAN through Parallel Computing

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- Observations



Introduction



- ▶ ADCIRC is an advanced three-dimensional circulation model and a finite element numerical model for hydrodynamic circulation
- ▶ designed for high computational efficiency, accuracy and numerical stability
- ▶ provides tidal and storm related hydrodynamic forcing for simulating storm surge phenomena



- ▶ ADCIRC is an advanced three-dimensional circulation model and a finite element numerical model for hydrodynamic circulation
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- ▶ provides tidal and storm related hydrodynamic forcing for simulating storm surge phenomena
- ▶ method: additive noise was introduced on body force

ADCIRC uses shallow water form of momentum equations coupled with continuity equation in spherical coordinates:

$$\begin{aligned}\frac{\partial u}{\partial t} + u S_p \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} - f v &= -g S_p \frac{\partial [\zeta + p_s / g \rho_0 - \alpha \eta]}{\partial x} + \frac{\partial}{\partial z} \left(\frac{\tau_{zx}}{\rho_0} \right) - b_x + m_x \\ \frac{\partial v}{\partial t} + u S_p \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} + f u &= -g \frac{\partial [\zeta + p_s / g \rho_0 - \alpha \eta]}{\partial y} + \frac{\partial}{\partial z} \left(\frac{\tau_{zy}}{\rho_0} \right) - b_y + m_y \\ \nabla \cdot \vec{u} &= S_p \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0\end{aligned}$$



Numerical Formulation

Variational Formulation

inner product wrt horizontal and vertical direction



$$\begin{aligned}
 & \left\langle \left\langle \frac{\partial u}{\partial t}, \phi_j \right\rangle_{\Omega}, \psi_k \right\rangle_Z + \left\langle \left\langle \left(u \frac{\partial u}{\partial x_{\sigma}} + v \frac{\partial u}{\partial y_{\sigma}} + w_{\sigma} \left(\frac{a-b}{H} \right) \frac{\partial u}{\partial \sigma} \right), \phi_j \right\rangle_{\Omega}, \psi_k \right\rangle_Z - \left\langle \left\langle f v, \phi_j \right\rangle_{\Omega}, \psi_k \right\rangle_Z = \\
 & - \left\langle \left\langle g \frac{\partial [\zeta + P_s / g \rho_o - \alpha \eta]}{\partial x}, \phi_j \right\rangle_{\Omega}, \psi_k \right\rangle_Z + \left\langle \left\langle \left(\frac{a-b}{H} \right) \frac{\partial}{\partial \sigma} \left(\frac{\tau_{zx}}{\rho_o} \right), \phi_j \right\rangle_{\Omega}, \psi_k \right\rangle_Z \\
 & - \left\langle \left\langle b_x, \phi_j \right\rangle_{\Omega}, \psi_k \right\rangle_Z + \left\langle \left\langle m_{x\sigma}, \phi_j \right\rangle_{\Omega}, \psi_k \right\rangle_Z \\
 \\
 & \left\langle \left\langle \frac{\partial v}{\partial t}, \phi_j \right\rangle_{\Omega}, \psi_k \right\rangle_Z + \left\langle \left\langle \left(u \frac{\partial v}{\partial x_{\sigma}} + v \frac{\partial v}{\partial y_{\sigma}} + w_{\sigma} \left(\frac{a-b}{H} \right) \frac{\partial v}{\partial \sigma} \right), \phi_j \right\rangle_{\Omega}, \psi_k \right\rangle_Z + \left\langle \left\langle f u, \phi_j \right\rangle_{\Omega}, \psi_k \right\rangle_Z = \\
 & - \left\langle \left\langle g \frac{\partial [\zeta + P_s / g \rho_o - \alpha \eta]}{\partial y}, \phi_j \right\rangle_{\Omega}, \psi_k \right\rangle_Z + \left\langle \left\langle \left(\frac{a-b}{H} \right) \frac{\partial}{\partial \sigma} \left(\frac{\tau_{zy}}{\rho_o} \right), \phi_j \right\rangle_{\Omega}, \psi_k \right\rangle_Z \\
 & - \left\langle \left\langle b_y, \phi_j \right\rangle_{\Omega}, \psi_k \right\rangle_Z + \left\langle \left\langle m_{y\sigma}, \phi_j \right\rangle_{\Omega}, \psi_k \right\rangle_Z
 \end{aligned}$$

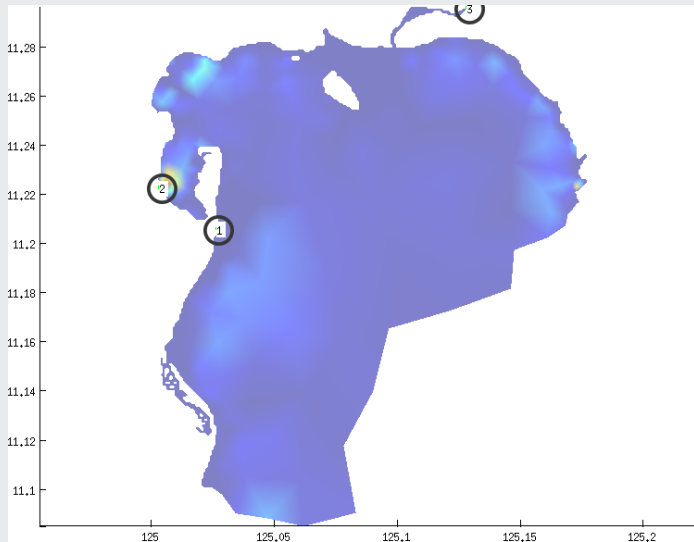
$$\begin{aligned}
 & (1 + i\Delta t f \alpha_1) \sum_{m=1}^3 q_{j,k+m-2}^{s+1} Inm_{k,m} + \alpha_3 \Delta t \left(\frac{a-b}{H_j^{s+1}} \right)^2 \sum_{m=1}^3 q_{j,k+m-2}^{s+1} KVnm_{j,k,m}^s \\
 & + \left(\frac{(a-b)\Delta t \alpha_2 K_{slip j}^s}{H_j^{s+1}} \right) q_j^{s+1} \Big|_{k=1} \\
 & = \sum_{m=1}^3 \left\{ (1 - i\Delta t f (1 - \alpha_1)) q_{j,k+m-2}^s - \Delta t [ladvec + lstress + bcp g]_{j,k+m-2} \right\} Inm_{k,m} \\
 & - \Delta t vadvec_{j,k} - \Delta t btpg_j LVn_k - (1 - \alpha_3) \Delta t vstress_{j,k}^s - \left(\frac{(a-b)\Delta t (1 - \alpha_2) K_{slip j}^s}{H_j^s} \right) q_j^s \Big|_{k=1} \\
 & + \frac{\Delta t (a-b)}{2} \left[\frac{\tau_{sxj}^{s+1}}{H_j^{s+1} \rho_o} + \frac{\tau_{sxj}^s}{H_j^s \rho_o} + i \left(\frac{\tau_{syj}^{s+1}}{H_j^{s+1} \rho_o} + \frac{\tau_{syj}^s}{H_j^s \rho_o} \right) \right]_{k=NV}
 \end{aligned}$$

Reduces to tridiagonal linear system $Mq = F$, solved by Jacobi conjugate gradient method.



Methodology

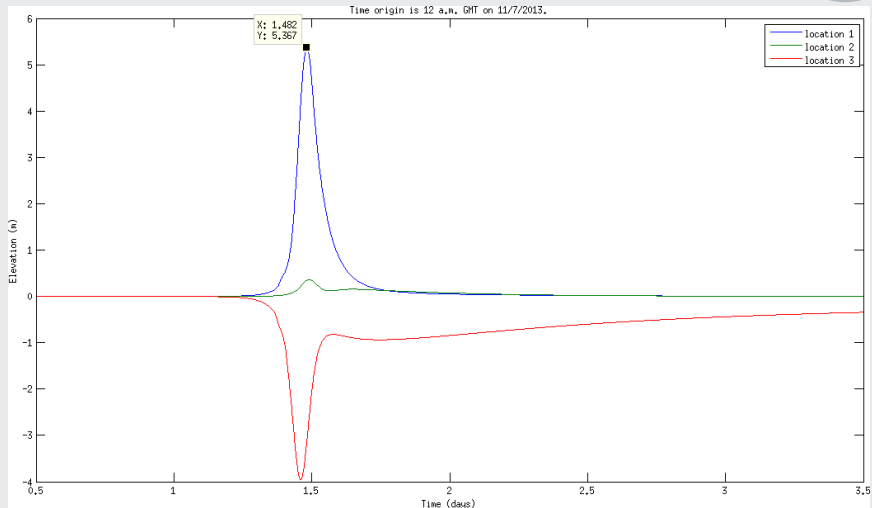
- ▶ Noise was added to force vector $F \leftarrow F + \xi$ to simulate random external force
- ▶ $\xi_1 \sim \mathcal{U}(0, 10^{-4})$ and $\xi_2 \sim \mathcal{U}(0, 10^{-3})$
- ▶ Modified source code to include additive noise
- ▶ Measured elevation on three points: near Tacloban airport (1), near Tacloban main (2) and an outstanding location (3)
- ▶ Compared *stochastic* results with *deterministic* result



Results

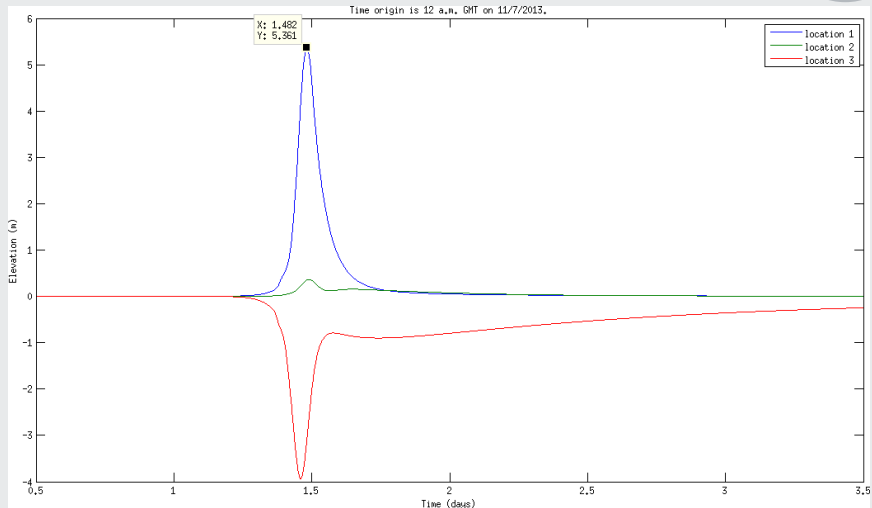
Time-series Elevation

deterministic: $\xi = 0$



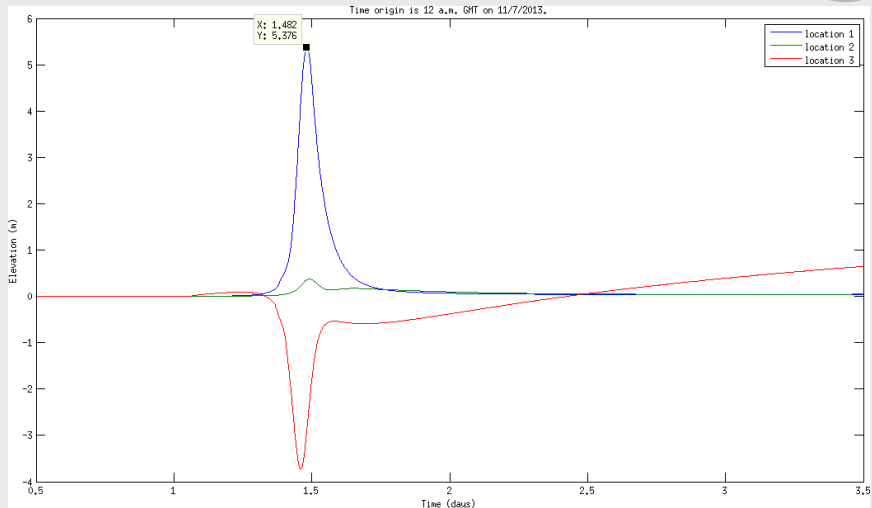
Time-series Elevation

stochastic: $\xi_1 \sim \mathcal{U}(0, 10^{-4})$



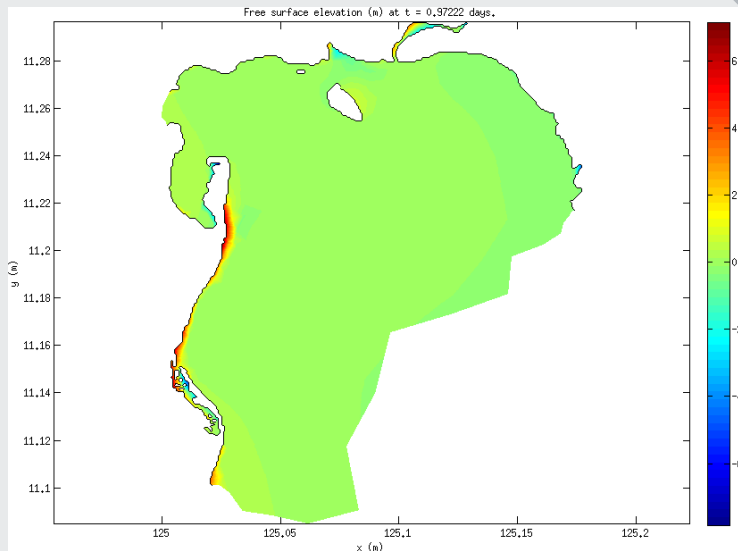
Time-series Elevation

stochastic: $\xi_2 \sim \mathcal{U}(0, 10^{-3})$



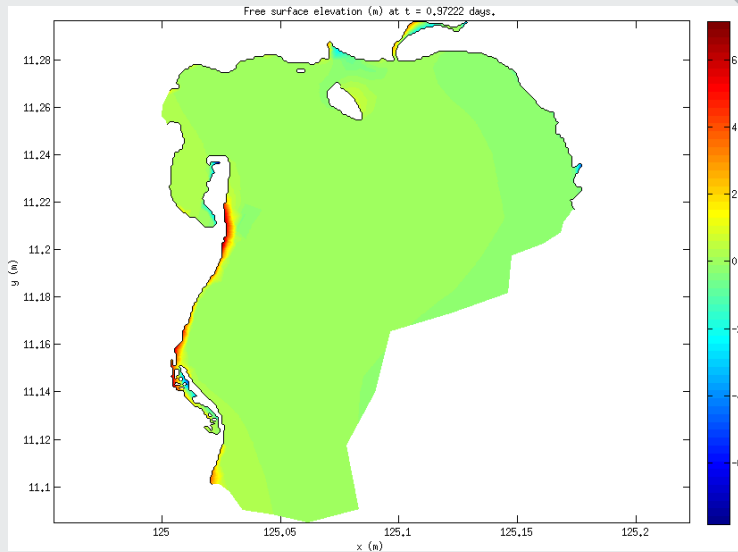
Surge Height on Domain

deterministic: $\xi = 0$



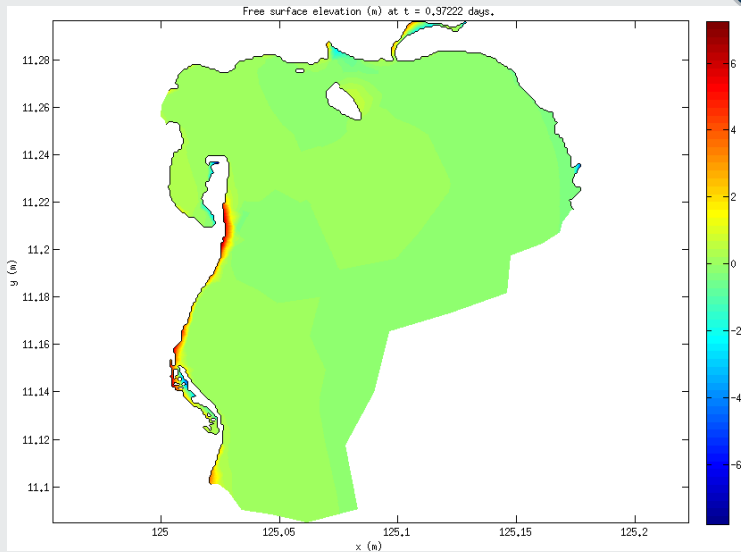
Surge Height on Domain

stochastic: $\xi_1 \sim \mathcal{U}(0, 10^{-4})$



Surge Height on Domain

stochastic: $\xi_2 \sim \mathcal{U}(0, 10^{-3})$



Storm Surge Simulation

deterministic: $\xi = 0$



Storm Surge Simulation

stochastic: $\xi_1 \sim \mathcal{U}(0, 10^{-4})$



Storm Surge Simulation

stochastic: $\xi_2 \sim \mathcal{U}(0, 10^{-3})$



Storm Surge Simulation

side-by-side comparison



- ▶ Time-series elevation curves still smooth because of the predictor-corrector algorithm employed by ADCIRC
- ▶ Surface elevation contours still show coherent structures despite noise
- ▶ Model is very sensitive to noise and larger perturbations lead to unstable computation
- ▶ This explains usage of uniform distribution rather than colored Levy process



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US Army Engineer Waterways Experiment Station, Vicksburg, Mississippi, 1992.



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Formulation and Numerical implementation of the 2D/3D ADCIRC Finite Element Model

<http://adcirc.org>, 2004.

Thank you for your attention!

